

35. Morphological Black Hat

Aim

To perform morphological Black Hat operation on an image using OpenCV.

Procedure

1. Read the image.
2. Convert it to grayscale.
3. Create a kernel.
4. Apply Black Hat transformation.
5. Display the output.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.ones((5, 5), np.uint8)

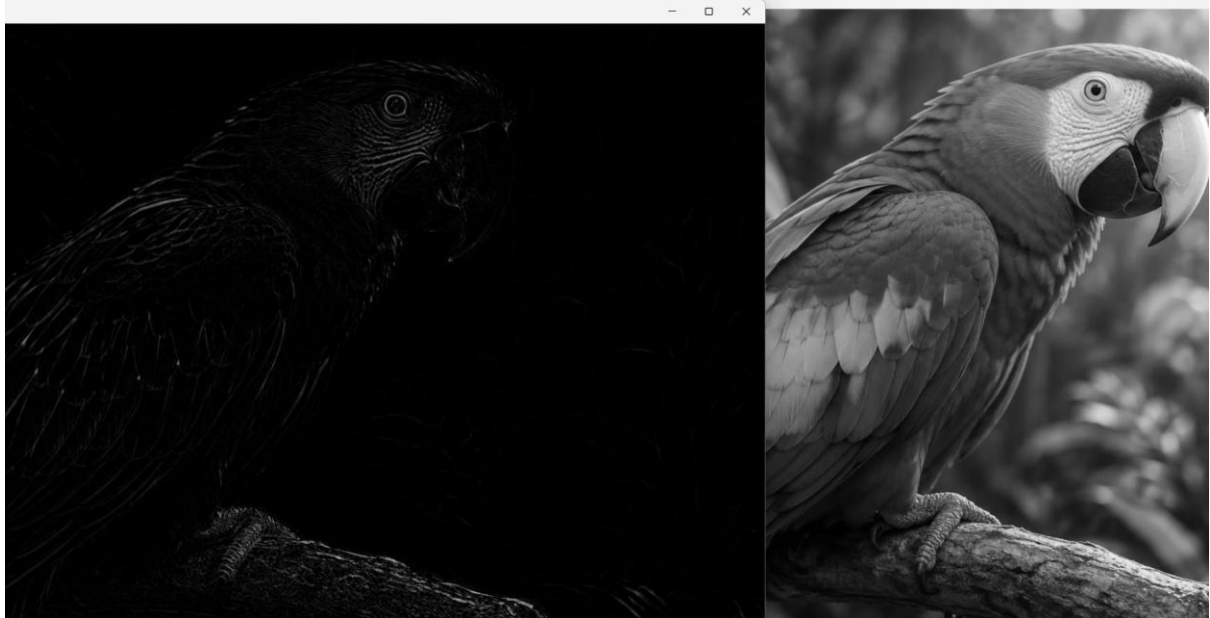
black_hat = cv2.morphologyEx(
    gray,
    cv2.MORPH_BLACKHAT,
    kernel
)

cv2.imshow("Original", gray)
cv2.imshow("Black Hat", black_hat)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

output



Result

The Black Hat morphological transformation was successfully performed on the given image.